Tirosh 2016 Melanoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → **GO**

6 samples (≥182 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached); mean cross-sample kNN fraction 0.47. All cells MITF-high — a single melanocytic compartment carries the trajectory (no lobe split).

Gate 2 — module geography → **GO** (non-branching)

Gast/pSom co-vary (r=+0.93) and decline along pseudotime (ρ=-0.73/-0.71); Crest anti-correlated (r=-0.81/-0.83), rising to a terminal readout (ρ=+0.41; crest-high decile mean pt=0.92).

Gate 3 — hub mutual exclusivity → **NA**

Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

Gate 1: integration

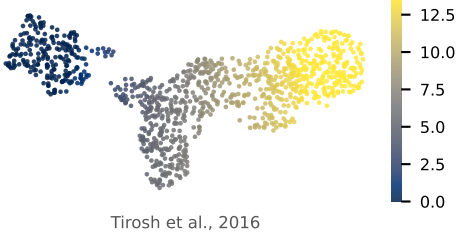
Melanoma manifold — samples



Tirosh et al., 2016

Trajectory / root

Pseudotime



Tirosh et al., 2016

Gast

Gastrulation Signature



Tirosh et al., 2016

pSom

Somitogenesis Signature



Tirosh et al., 2016

Crest

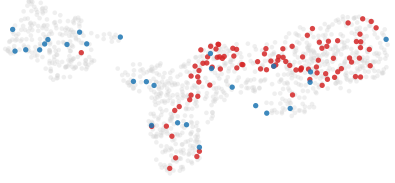
Neural Crest Signature



Tirosh et al., 2016

EMT-TFs

TWIST1 (early) · SNAI2 (late)



Tirosh et al., 2016

Dataset.

Melanoma (Tirosh et al., 2016), SmartSeq2 — 19 patients / 4,645 cells (master scope); malignant only: 1,257 load across 15 samples, 1,061 across 6 samples (≥182 cells) enter preprocessing. All cells form one continuous MITF-high melanocytic compartment that carries the trajectory (no lobe split).

Verdict.

✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Tirosh's melanoma manifold is a single monotone Gast/pSom → terminal-Crest gradient (non-branching), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. Data-derived EMT-TFs are TWIST1 (early/root, ρ=-0.05) → SNAI2 (late/terminus, ρ=+0.44); Puram's canonical SNAI1/ZEB1 are expressed (10%/11%) but not the most end-specific factors here.

Parameter choices

Min cells/sample	≥ 182 (keeps 6/15; gap 98→54)	**Integration**	Harmony(sample) θ=2; PCA 50PC; UMAP spread=1.0
Compartment	single melanocytic (all MITF-high)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip = Gast/pSom-high, Crest-low	**Subtypes**	Cycling/Hypoxic, Melanocytic-quiescent, Pigmented-Stress